Computer Networks Lab (BCSE308P)

Fall Semester 2026-27

Smart Campus NOC Simulator 2.0

"Design and Performance Analysis of a Secure, Fault-Tolerant Smart Campus Network"

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B.Tech.

in

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PROJECT DESCRIPTION AND OUTPUT

1. PROJECT OVERVIEW

1.1 Introduction

Smart Campus Network is a browser-based simulation platform for a university campus network. It models IP addressing, VLANs, routing (OSPF and RIP), redundancy and failover, ACL-based security, QoS, network services and data-link protocols entirely in software, without physical routers or switches. All traffic, delay and packet-loss figures are simulator outputs.

1.2 Architecture

Table 1. Architecture of the project

Part	Technology	Role
Frontend	React 18, TypeScript, Vite, Tailwind CSS, @xyflow/react, Recharts	Dark NOC-style web interface at http://localhost:5173
Backend	Node.js, Express, TypeScript, Vitest	API on port 5000; routing, ACL, QoS, packet-tracing, services and protocol engines
Database	sql.js (SQLite in WebAssembly)	Stores devices, links, VLANs, ACL rules and experiment records
C programs	ANSI C (C99)	Nine standalone syllabus algorithms

1.3 Application pages

Table 2. Pages of the application

Page	Function	Concept
Dashboard	Availability, throughput, latency, security incidents and traffic charts	Network monitoring
Network Topology	Interactive device graph and hop-by-hop packet tracer	Packet forwarding
IP & VLAN Manager	Subnet calculator, VLSM planner, VLAN and IPv4/IPv6 plan	Addressing, VLANs
Routing Simulator	Shortest path using OSPF, RIP or static routes	Routing algorithms
Fault Tolerance Lab	Link failure with convergence time and packet loss	Redundancy, convergence
Security Center	Ordered ACL rule chain and packet inspection tester	Firewalls, segmentation
Traffic & QoS Analyzer	FIFO versus strict-priority queueing under congestion	Congestion, QoS
Network Services	DHCP, DNS, HTTP, FTP and NAT/PAT models	Application services
Protocol Lab	Hamming, CRC, checksum, ARQ, ALOHA, CSMA, CSMA/CD	Data-link layer

2. THE SIMULATED CAMPUS NETWORK

2.1 Topology

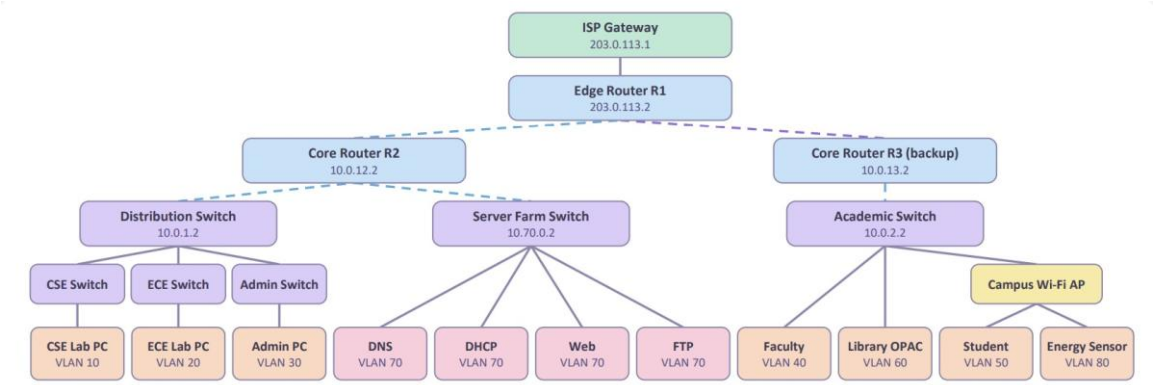


Figure 1. Campus network topology: ISP gateway, edge router R1, core routers R2 and R3 (backup), switches, servers and end devices with their VLANs.

The network has 22 devices and 22 links. Edge router R1 connects to core router R2 (primary, cost 5) and core router R3 (backup, cost 12), and R2 and R3 are joined by a trunk (cost 5). This triangle provides the alternate path used in the fault-tolerance experiments. Departmental switches, servers (DNS, DHCP, web, FTP), the Wi-Fi access point and end devices hang below the core, each in its own VLAN.

2.2 VLANs and addressing

Nine VLANs separate the campus into broadcast domains, each with a /24 IPv4 subnet, a gateway at .1 and a matching IPv6 /64 prefix.

Table 3. VLAN plan of the campus

VLAN	Name	IPv4 subnet	Devices
10	CSE Department	10.10.0.0/24	CSE lab PCs
20	ECE Department	10.20.0.0/24	ECE hardware lab
30	Administration	10.30.0.0/24	Finance, registrar, exam cell
40	Faculty / Staff	10.40.0.0/24	Faculty chambers
50	Student Network	10.50.0.0/24	Student Wi-Fi and BYOD
60	Central Library	10.60.0.0/24	OPAC terminals
70	Data Center Servers	10.70.0.0/24	Web, DNS, DHCP, FTP servers
80	IoT Sensors	10.80.0.0/24	Smart-energy sensors
99	Network Management	10.99.0.0/24	Router and switch administration

3. MODULE DESCRIPTIONS WITH SCREENSHOTS

3.1 Dashboard

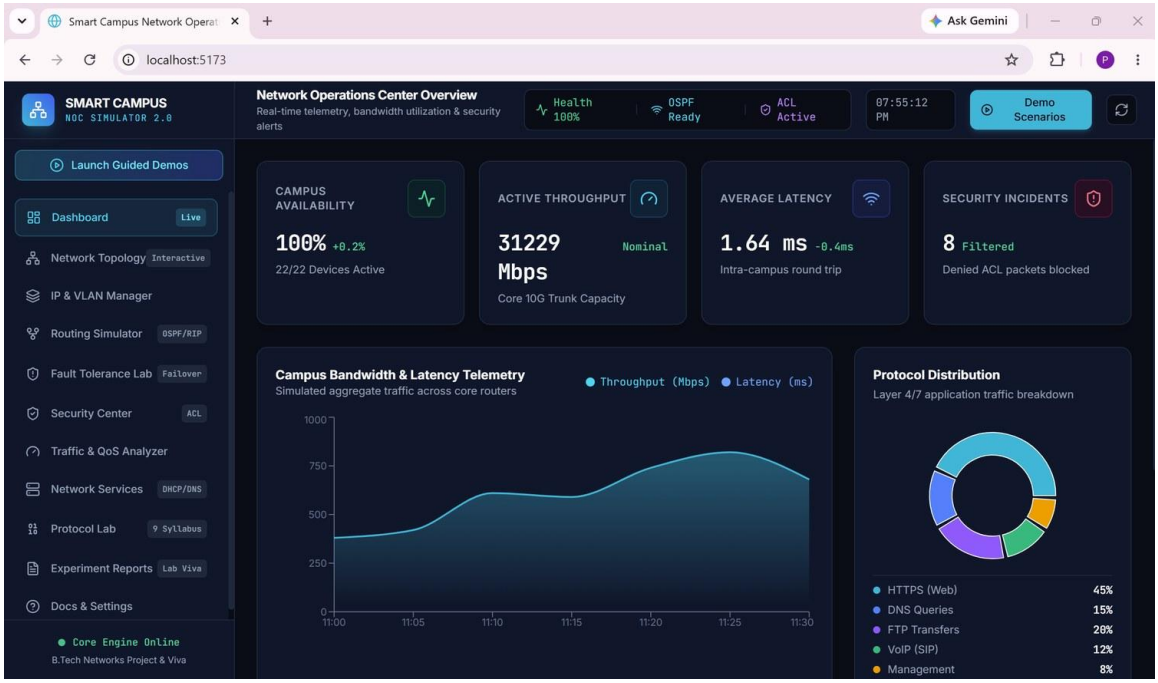


Figure 2. Dashboard: availability, throughput, latency, security incidents, telemetry and protocol distribution.

What it shows. Campus availability 100% (22/22 devices active), throughput on the 10G core, average latency 1.64 ms, 8 ACL-denied packets, a bandwidth/latency chart and the protocol mix (HTTPS 45%, FTP 20%, DNS 15%, VoIP 12%, Management 8%).

How it works. Device and link status come from the backend and are combined with simulated traffic data.

3.2 Network Topology and Packet Tracer

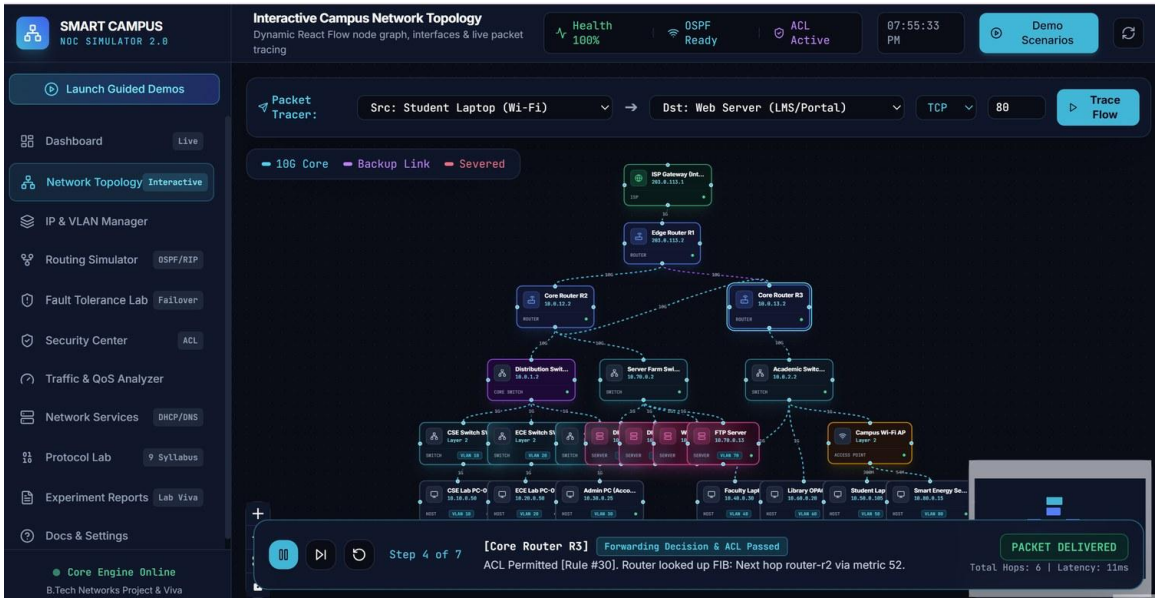


Figure 3. Packet tracer: TCP/80 from a student laptop to the LMS web server, delivered in 6 hops (11 ms).

What it shows. The interactive device graph with colour-coded links, and a tracer that follows a packet from a chosen source to a destination. Step 4 of 7 is Core Router R3, where the forwarding decision and ACL check pass (rule #30). The trace ends with PACKET DELIVERED, 6 hops, 11 ms.

How it works. The backend computes the route and, at each device, records the next hop and metric and, where an ACL applies, the matching rule and the allow or drop reason.

3.3 IP and VLAN Manager

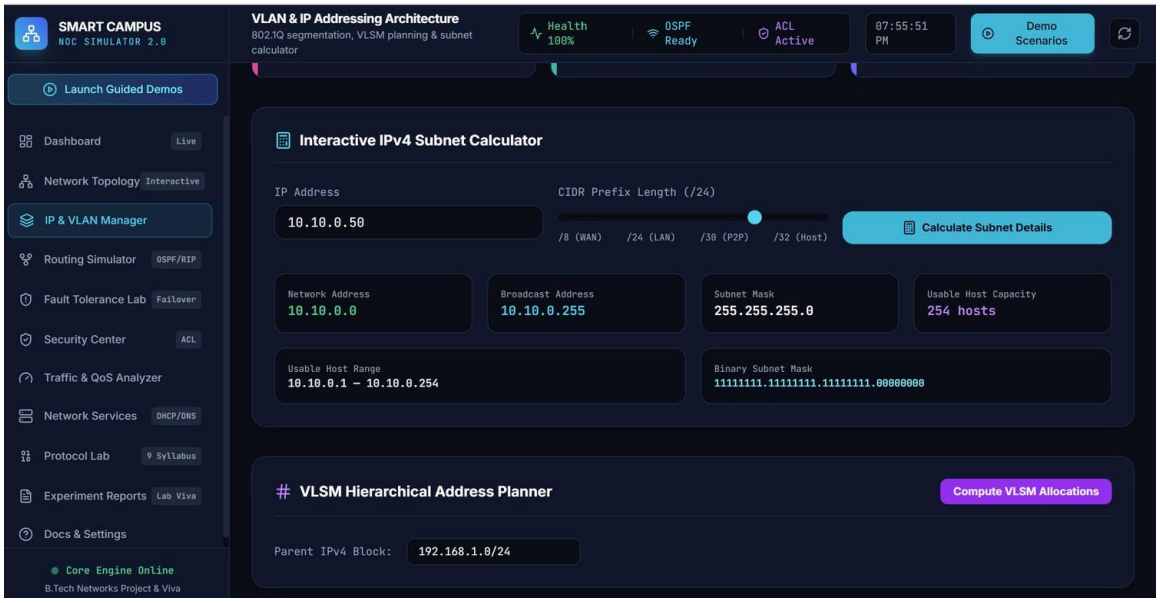


Figure 4. Subnet calculator: 10.10.0.50/24 gives network 10.10.0.0, broadcast 10.10.0.255 and 254 usable hosts.

What it shows. Network and broadcast address, subnet mask, usable host range, host capacity and binary mask for any IP and prefix length.

How it works. Network = IP AND mask; broadcast = network with all host bits set; usable hosts = 2^(host bits) - 2.

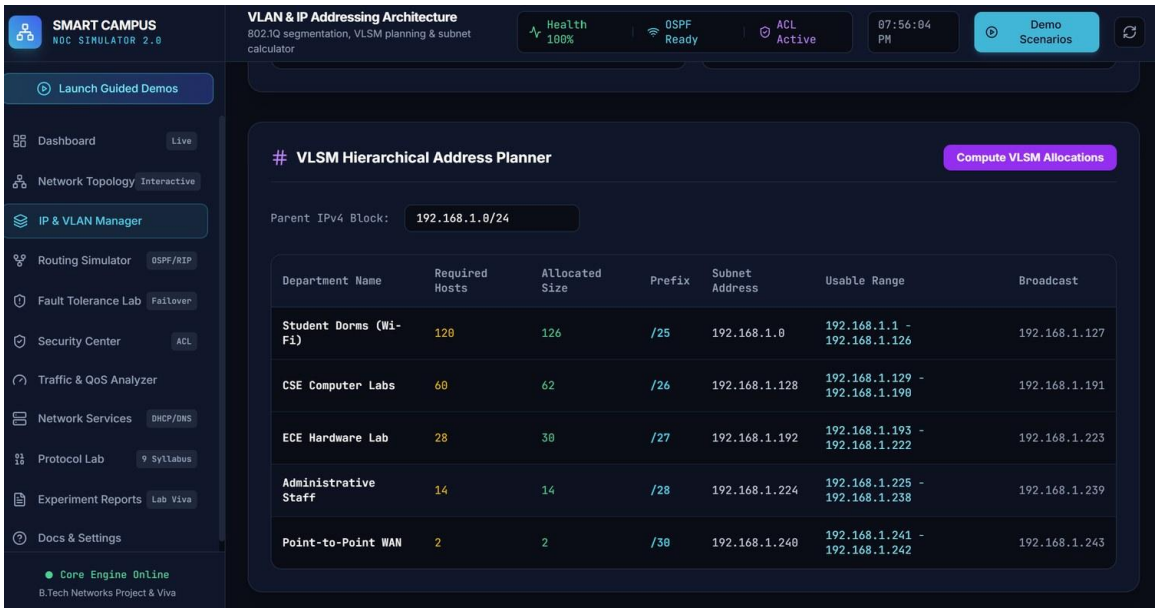


Figure 5. VLSM planner: 192.168.1.0/24 divided into /25, /26, /27, /28 and /30 subnets.

What it shows. Student Dorms (120 hosts) get 192.168.1.0/25, CSE Labs (60) get .128/26, ECE Hardware Lab (28) get .192/27, Administrative Staff (14) get .224/28 and the point-to-point WAN link (2) gets .240/30, with no overlap.

How it works. Departments are sorted by size and each receives the smallest subnet that fits, allocated one after another.

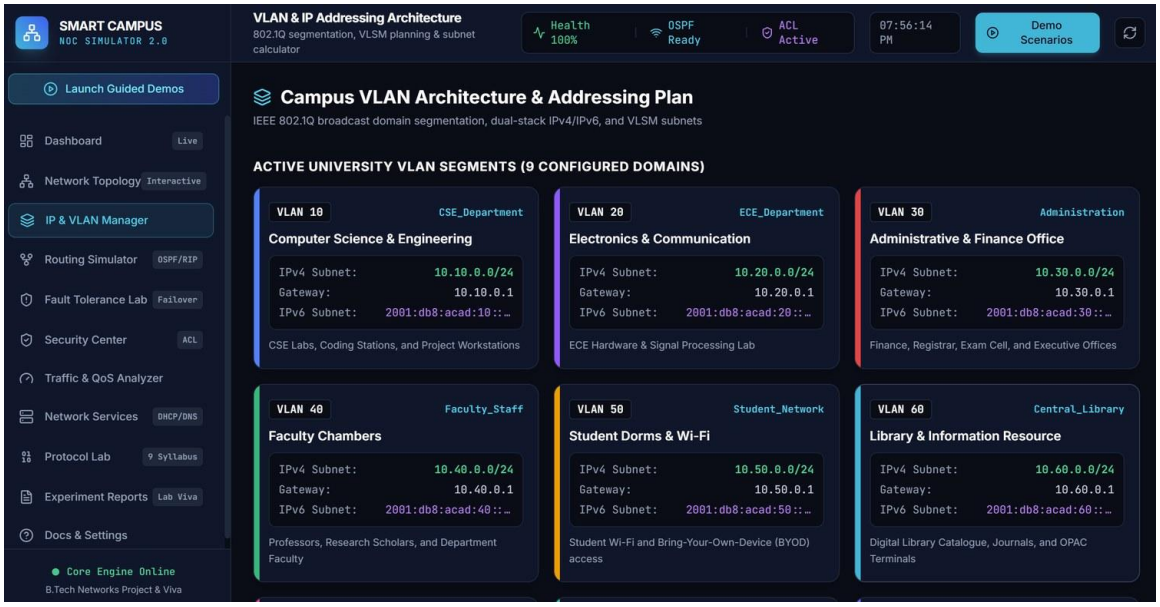


Figure 6. VLAN addressing plan: nine configured VLANs with IPv4 subnets, gateways and IPv6 prefixes.

What it shows. One card per VLAN with its department, IPv4 /24 subnet, gateway and IPv6 /64 prefix (IEEE 802.1Q segmentation).

3.4 Routing Simulator

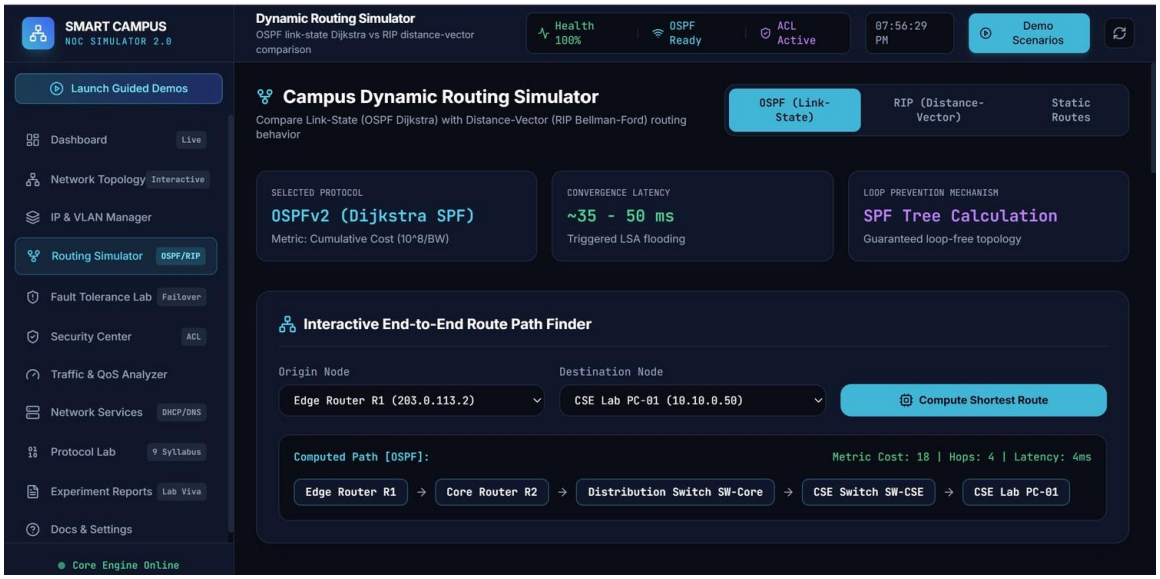


Figure 7. Routing simulator (OSPF): shortest path from Edge Router R1 to CSE Lab PC-01, cost 18, 4 hops, 4 ms.

What it shows. Tabs for OSPF, RIP and static routes. For OSPF the page shows the metric (10^8 / bandwidth), typical convergence of about 35 to 50 ms and the SPF-tree loop prevention. The computed path is R1, R2, SW-Core, SW-CSE, CSE Lab PC-01.

How it works. OSPF mode runs Dijkstra on the link-cost graph. RIP mode runs Bellman-Ford on hop counts, with 16 hops treated as unreachable.

3.5 Fault Tolerance Lab

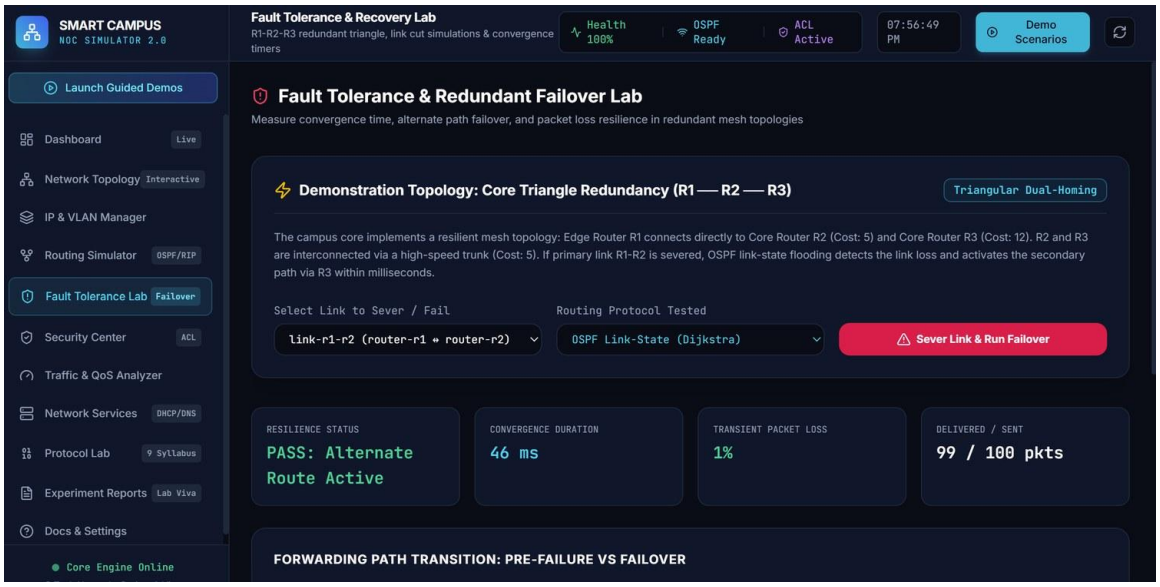


Figure 8. Fault tolerance (OSPF): link-r1-r2 severed; 46 ms convergence, 1% loss, 99/100 packets delivered.

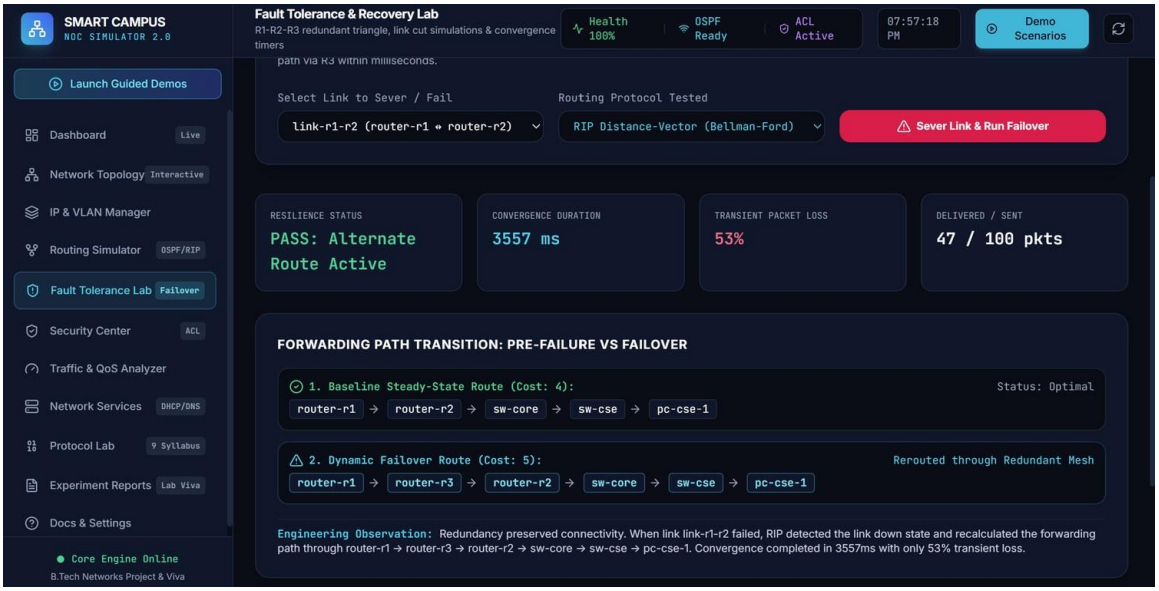


Figure 9. Fault tolerance (RIP): same failure; 3,557 ms convergence, 53% loss, 47/100 packets delivered.

What it shows. After the link is severed, traffic moves from R1-R2 to R1-R3-R2 under both protocols. OSPF converges in 46 ms with 1% loss; RIP takes 3,557 ms and loses 53% of the test packets.

How it works. The engine fails the chosen link on a copy of the topology, recomputes the route and estimates convergence and loss from the protocol model (OSPF-style 30 to 55 ms, RIP-style 2,400 to 3,600 ms). With no alternate path the destination is unreachable with 100% loss.

3.6 Security Center (ACLs)

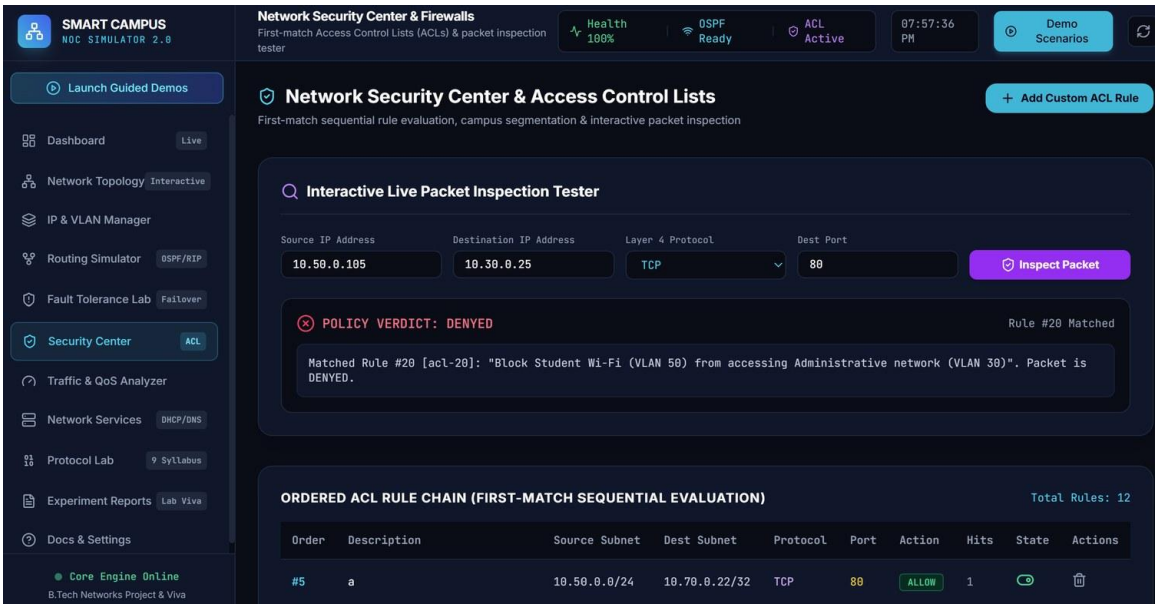


Figure 10. Packet tester: TCP/80 from 10.50.0.105 (Student VLAN 50) to 10.30.0.25 (Administration VLAN 30) is denied by rule #20.



Figure 11. Ordered ACL rule chain (12 rules) with hit counters.

What it shows. The tester returns DENIED for the student-to-administration packet, matched by rule #20. The rule chain lists deny rules for IoT (#10) and student (#20) traffic to Administration; allow rules for student web access to 10.70.0.12 on TCP 80 and 443 (#30, #31), DNS (#40) and the library (#50); SSH from the management subnet (#60) with all other SSH denied (#61); and ICMP inside 10.0.0.0/8 (#70).

How it works. Rules are evaluated in order and the first match decides; each match increments the rule's hit counter.

3.7 Traffic & QoS Analyzer

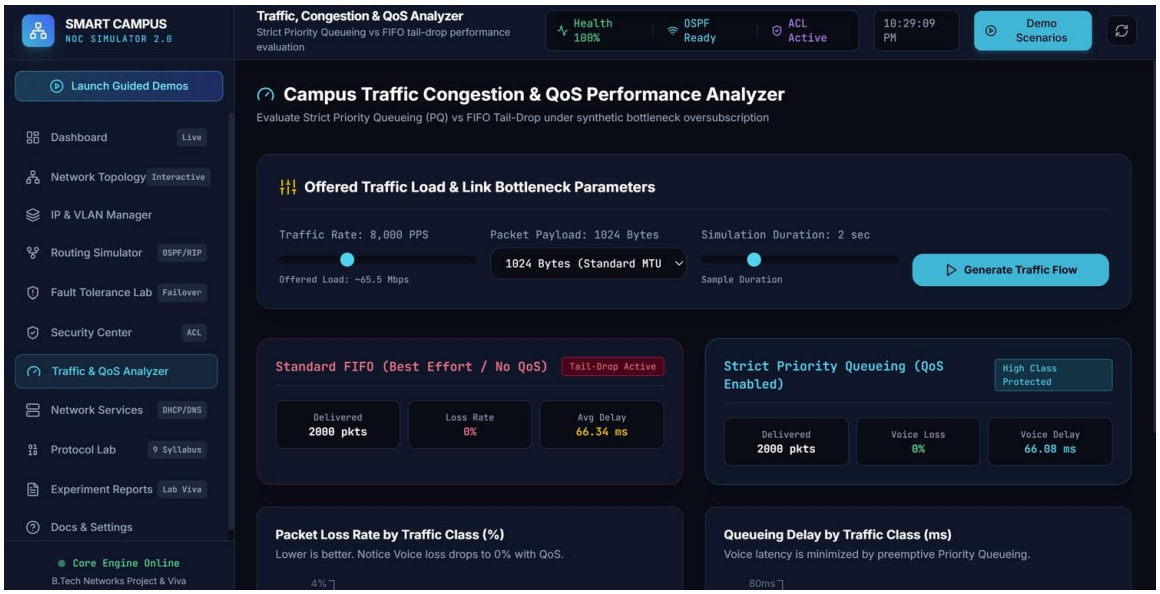


Figure 12. Traffic & QoS Analyzer: standard FIFO tail-drop versus strict-priority queueing for the same offered load.

What it shows. The offered load is set to 8,000 packets/s with a 1,024-byte payload for 2 seconds (about 65.5 Mbps). Under standard FIFO (best effort, tail-drop) 2,000 packets are delivered with 0% loss and a 66.34 ms average delay. Under strict-priority queueing 2,000 packets are delivered with 0% voice loss and a 66.08 ms voice delay. The charts below compare packet loss and queueing delay by traffic class.

How it works. One packet sequence is generated and run through both queue models. FIFO serves packets in arrival order and drops arrivals when the buffer is full; strict priority always serves the high-priority (voice) class first.

3.8 Network Services

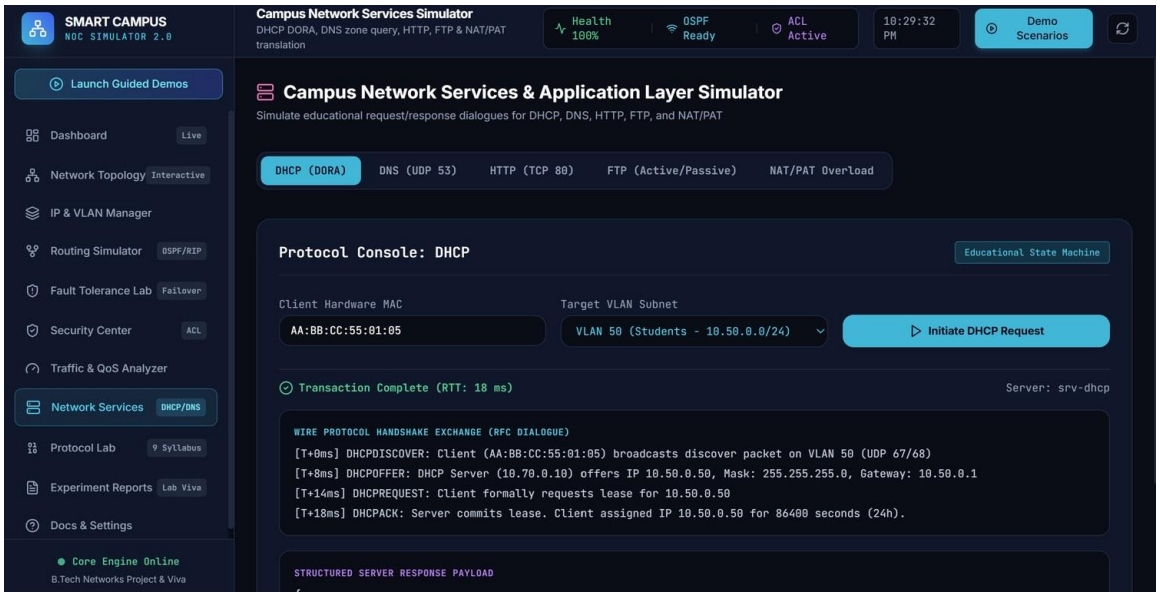


Figure 13. Network Services: DHCP (DORA) transaction for a client on VLAN 50.

What it shows. Tabs for DHCP, DNS (UDP 53), HTTP (TCP 80), FTP (active/passive) and NAT/PAT overload. In the DHCP console a client (MAC AA:BB:CC:55:01:05) on VLAN 50 completes the transaction in 18 ms: DISCOVER is broadcast on UDP 67/68 at T+0 ms, the server 10.70.0.10 sends an OFFER at T+8 ms (IP 10.50.0.50, mask 255.255.255.0, gateway 10.50.0.1), the client sends a REQUEST at T+14 ms, and the ACK at T+18 ms commits a 24-hour (86,400 s) lease.

How it works. Each protocol is an educational state machine that produces the timestamped request and response dialogue and a structured server response.

3.9 Protocol Lab

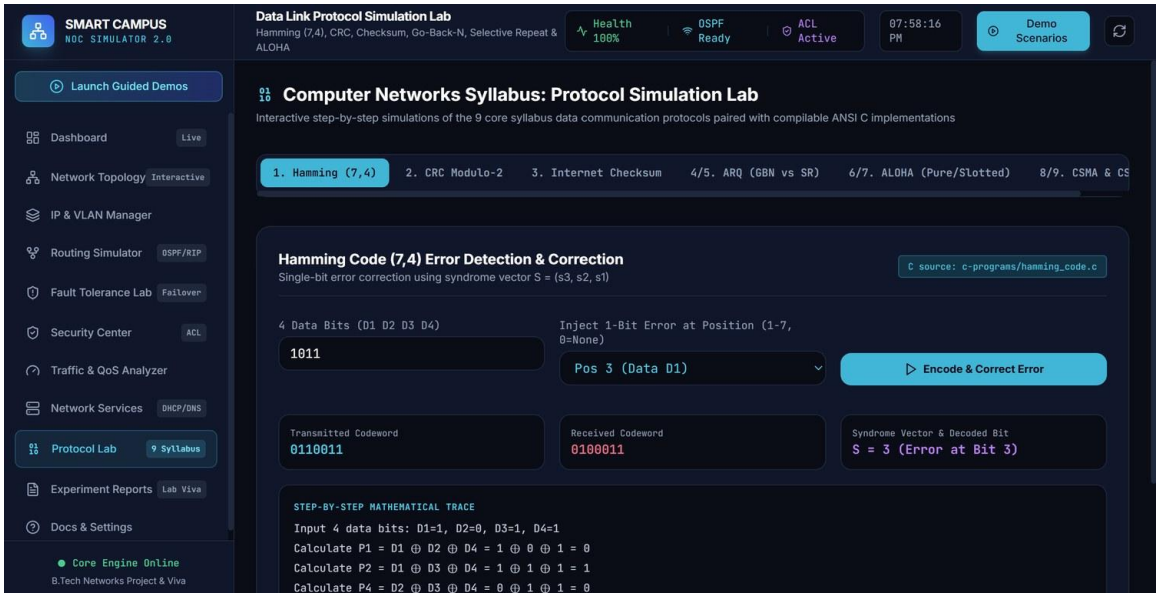


Figure 14. Hamming (7,4): data 1011 with an error at bit 3, located by syndrome S = 3 and corrected.

What it shows. Tabs for Hamming (7,4), CRC, Internet checksum, Go-Back-N, Selective Repeat, ALOHA and CSMA/CSMA-CD, each paired with a C program. Data 1011 is encoded as 0110011; with an error at position 3 the receiver gets 0100011, and the syndrome S = 3 identifies bit 3.

How it works. P1 = D1 xor D2 xor D4, P2 = D1 xor D3 xor D4, P4 = D2 xor D3 xor D4; the syndrome, read as a binary number, is the position of the faulty bit.

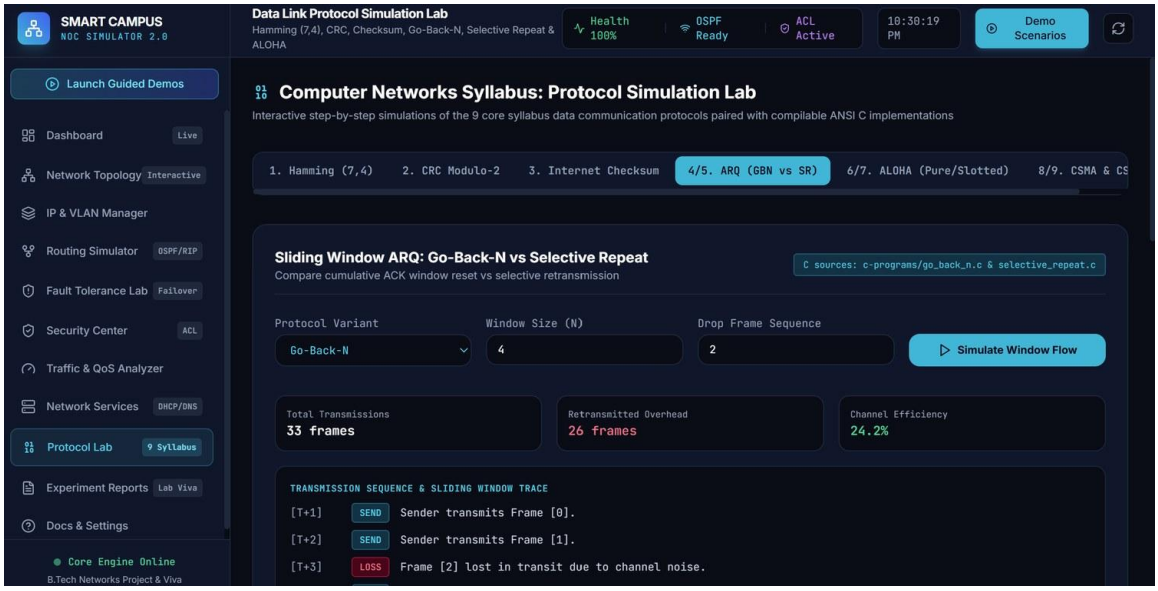


Figure 15. Sliding-window ARQ: Go-Back-N with window size 4 and frame 2 dropped.

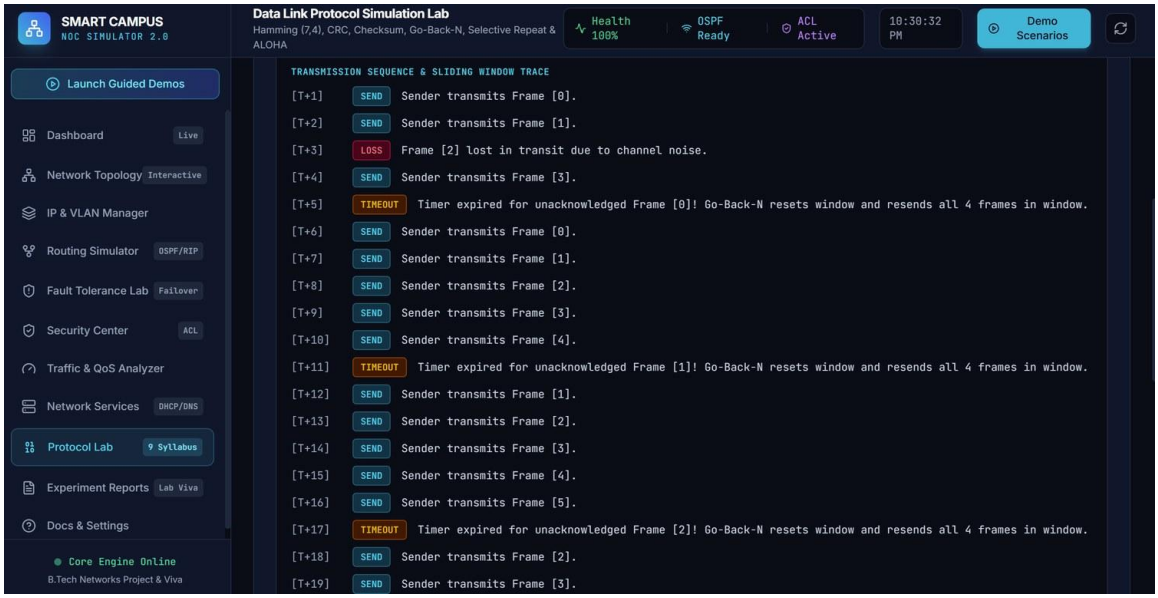


Figure 16. Go-Back-N transmission trace: after each loss the window is reset and all frames in it are resent.

What it shows. With window size 4 and frame 2 dropped, Go-Back-N makes 33 frame transmissions in total, of which 26 are retransmission overhead, giving a channel efficiency of 24.2%. In the trace, frames 0 and 1 are sent and frame 2 is lost; the timer for frame 0 then expires and Go-Back-N resets the window and resends all 4 frames in it. The same timeout and resend cycle repeats for frames 1 and 2.

How it works. Go-Back-N retransmits every frame from the first unacknowledged one, whereas Selective Repeat (selected in the same tab) retransmits only the lost frame, so Go-Back-N has the larger retransmission overhead.

4. DEMONSTRATION SCENARIOS

The application provides five guided scenarios (Launch Guided Demos). The steps below reproduce each one on the individual pages.

Scenario 1: Normal web request across VLANs

1. Open Network Topology; set source Student Laptop (Wi-Fi), destination Web Server, TCP, port 80.
2. Press Trace Flow and step through the trace (Figure 3).
3. Result: delivered in 6 hops, 11 ms; ACL rule #30 permits the traffic.

Scenario 2: Security ACL enforcement

1. Open Security Center; enter source 10.50.0.105, destination 10.30.0.25, TCP, port 80.
2. Press Inspect Packet. Result: DENIED by rule #20 (Figure 10).
3. Check the hit counter of rule #20 in the rule chain (Figure 11).

Scenario 3: Dynamic fault recovery

1. Open Fault Tolerance Lab; select link-r1-r2 and OSPF Link-State.
2. Press Sever Link & Run Failover. Result: 46 ms, 1% loss, 99/100 delivered (Figure 8).
3. Repeat with RIP. Result: 3,557 ms, 53% loss, 47/100 delivered (Figure 9).

Scenario 4: Congestion and QoS

1. Open Traffic & QoS Analyzer and press Generate Traffic Flow (Figure 12).
2. Compare FIFO with strict-priority queueing for the same packet stream.
3. Result: 2,000 packets delivered in both cases with 0% voice loss.

Scenario 5: Data-link error correction

1. Open Protocol Lab, tab Hamming (7,4); enter data bits 1011.
2. Select error position 3 and press Encode & Correct Error (Figure 14).
3. Result: transmitted 0110011, received 0100011, syndrome S = 3, error corrected.

5. EXPERIMENTS AND RESULTS

Each experiment was run once on the application with its default configuration; values are single-run simulator outputs.

Table 4. Experiments and observed results

No.	Experiment	Setup	Observed result
E1	Normal routing	OSPF, R1 to CSE Lab PC-01	Cost 18, 4 hops, 4 ms
E2	Packet trace	Student laptop to web server, TCP/80	Delivered, 6 hops, 11 ms, permitted by rule #30
E3	Core link failure, OSPF	Sever link-r1-r2, 100 packets	46 ms convergence, 1% loss, 99/100 delivered
E4	Core link failure, RIP	Sever link-r1-r2, 100 packets	3,557 ms convergence, 53% loss, 47/100 delivered
E5	Bottleneck QoS	FIFO vs strict priority; 8,000 pkt/s, 1,024 B, 2 s	Both deliver 2,000, 0% loss; 66.34 ms vs 66.08 ms
E6	ACL policy	Student 10.50.0.105 to Admin 10.30.0.25, TCP/80	Denied by rule #20 (9 hits); rule #10 has 0 hits
E7	No alternate path	Fail the only access link to a host	Unreachable, 100% simulated loss (automated test)
E8	Subnet and VLSM	10.10.0.50/24; 192.168.1.0/24 for 120/60/28/14/2 hosts	254 hosts; /25, /26, /27, /28, /30
E9	Hamming (7,4)	Data 1011, error at bit 3	Syndrome 3, error corrected
E10	DHCP (DORA)	Client AA:BB:CC:55:01:05 on VLAN 50	Lease 10.50.0.50 for 24 h; transaction complete in 18 ms
E11	Go-Back-N ARQ	Window size 4, frame 2 dropped	33 transmissions, 26 retransmitted, 24.2% efficiency

Automated verification. 23 backend tests (Vitest) and 22 HTTP/API checks pass, and both production builds succeed.